

AI-BASED MUSIC RECOMMENDATION SYSTEM

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Abstract

The need for such personalized music recommendation systems arises from the rapid expansion of digital music libraries and is an essential part of the modern streaming platforms. The focus of this research is about developing an AI-Based Music Recommendation System by using K-Means Clustering and K-Nearest Neighbors (KNN) algorithm to enhance user experience through precision and efficiency of song recommendations. In particular, the study attempts to tackle some critical issues like user preference modeling, computational efficiency, and real-time recommendation accuracy.

The approach performs data preprocessing, finds the features that can be extracted from data, clusters similar music tracks and uses KNN to give a personalized recommendation based on what the user has listened to. Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used to evaluate the model and obtained low error values in both training and the testing datasets. To test the system, I input a sample song, 'All the Time - Don Diablo Remix' and obtained 5 other similar tracks, thus validating the system.

To this end, integrating machine learning techniques into AI driven music recommendation systems, this research contributes to these systems with the improvement to accuracy, efficiency, and user satisfaction. Further work will include addition of deep learning models, real time learning mechanism and data integrations of multi modal data to further optimize the music recommendations. This study's findings indicate the possibilities of how AI can transform the user's experience of personalized music, making digital music use more engaging and user centric.

Keywords: AI-Based Music Recommendation, K-Means Clustering, K-Nearest Neighbors (KNN), Machine Learning, Music Personalization, User Experience, Recommendation Systems.

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It is important to have a special appreciation to Spotify and other public sources of music data in order to be able to analyze and even develop models based on the released data. For that, I am grateful to the open-source machine learning and data science communities for the documentation, research papers and programming frameworks that greatly assisted in the development of this system.

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1 Introduction

Music is the art of arranging sound using different components like melody, harmony, rhythm, and timbre in time to produce an experience that is expressive or significant. Nowadays, users gather a large scale of digital music in their digital devices, but the problem is to manage them according to their taste. It is a difficult process to manage the songs that they need to listen in their daily lives, without using pre-determined organization such as albums, playlists or computationally generated recommendations. So, a good recommendation system is required which can be able to minimize user's effort required to provide feedback and simultaneously to maximize the user's satisfaction (Hu and Ogihara, 2011).

Recently, there has been a lot of interest in music recommendation systems that use machine learning and deep learning algorithms to analyse listener data and optimize their satisfaction level. There are different types of models that nowadays apply to make the best music recommendation system for their users (Karishma, 2024). Different music platforms like Amazon Music, Soundcloud and Spotify also utilises the advanced music recommendation systems based on artificial intelligence to enhance the music recommendation systems for their users.

Digital music has become a basic part of our daily life and users store huge music collections across their connected devices. People find it hard to organize their favourite music when they depend on their own selections alone. The development of AI-based music recommendation systems uses machine learning and deep learning to examine listener data and improve user experiences. Leading music platforms such as Amazon Music and Spotify enhance recommendation systems through AI algorithms to find suitable music matches for their users. This project aims to develop an intelligent music recommendation system utilizing machine learning approaches, particularly K-Means clustering and the K-Nearest Neighbours (KNN) model, to optimize user satisfaction and minimize manual effort in music selection.

1.1 Aims

- An AI-based Music Recommendation System develops to provide improved user experiences through predictions regarding user song preferences.
- The design of K-Means clustering to organize songs through musical features will boost recommendation precision.
- The application of K-Nearest Neighbours (KNN) algorithm analyses user listening behaviours to enhance the refinement of personalized recommendations.
- This system will help reduce human workload because machine learning techniques will automatically produce song recommendations.
- The research aims to compare how well clustering and classification models enhance recommendation accuracy while influencing user satisfaction through evaluation techniques.

1.2 Objectives

- To build AI- Based Music Recommendation System.
- To obtain the K-means cluster analysis for the Music Recommendation System.
- To apply the K-Nearest Neighbours (KNN) model to enhance the Music Recommendation System.

1.3 Research gap

While much has been achieved in AI-driven music recommendation systems, there is still work to be done both in around the existing models.

Less Personalization Accuracy — many existing music recommendation systems have been built on collaborative filtering or content-based filtering, which often cannot accurately reflect the users' dynamic personal preferences and contextual listening behaviour. To enable this, there is a need for a more adaptive model that combines several approaches that include clustering techniques and classification techniques.

Limited Use of Hybrid Approaches – Most commonly machine learning algorithms like KMeans and KNN are used to solve many problems in different domains but have not been widely explored to solve the problem of content recommendation for Music. Almost all systems already existing either consistently lie on the side of deep learning or consider neural networks, neglecting effectiveness of the traditional methods of clustering and classification. (Yang, 2023)

Performance issues with recommendation systems arise from their limitations to efficiently process large music datasets efficiently. The solution for creating real-time recommendations along with effective system performance remains an unaddressed problem. (Yang, 2023)

Contextual Appropriateness – The current crop of music recommending systems do not take context awareness into account for instance mood, activity or location related preferences. Context aware filtering techniques can have good effect on user experience.

Diversity and Novelty Issues —many recommendation systems are that they tend to reproduce users' existing choices of music, which produces the so called 'filter bubble', by pushing familiar genres. Personalized recommendations that also include musical reinforcement for diversity is essential for models that also need to deliver musical diversity that goes beyond a user's general listening patterns.

Addressing Bias and Ensuring Fairness: Recommendation algorithms are not perfect, and they can develop biases; for instance, the algorithm may favour known artists and therefore give minority a raw deal when it comes to artist recommendation. That is why carried out for the purpose of creating non-biased methods for music recommendation system are essential (Knibbe, 2021).

Enhancing Interpretability and Transparency: Most of the existing recommendation systems are black boxes therefore users cannot explain why a particular recommendation has been made. This work was done to fill the gap of understanding explainable AI techniques for the

development of trust and the ability of users to comprehend how the solutions are recommended (Murindanyi et al., 2024).

1.4 Ethical considerations

AI has raised several ethical concerns in case of Music Recommendation System, and they include the fairness, privacy, and responsible AI usage. The following are the details of the ethical consideration:

1.4.1 User Privacy and Data Protection

- Specifically, the system must also maintain user data compliance in various regulatory principalities including GDPR rules.
- Users should be informed how their listening history and the data is utilized for recommendation and other purposes (Zhou et al., 2023).
- Appropriate measures of masking and securing the collected data reduces likelihood of leakage or misuse of the information.

1.4.2 Avoiding Manipulative Algorithms

- Popular music streaming services show more sponsored content and commercially useful suggestions instead of content that matches user preferences.
- The system needs to follow ethical advertising standards by delivering genuine personalized recommendations instead of recommending products for financial gain (Yang, 2023).

1.4.3 Psychological and Social Impact

- Customized recommendation systems have the potential to restrict users from experiencing different types of music because they show content based on personal preferences.
- Users should experience a balance between personalized recommendations and discovery of new music through the system (Knibbe, 2021).

2 Literature review

In this section, a comprehensive summary will be explained regarding the recommendation of a music-based system of past studies. Also, the techniques used and their results or achievements for the Music Recommendation System will be explained. The old and previous studies used different methods to build an automatic music recommendation system using various types of music datasets. Artificial intelligence (AI) plays an important role nowadays in enhancing performance and improving the efficiency, automation, and decision-making process. The AI-based models also gained significant popularity in different streaming services. These models provide facilities for an automatic music recommendation system for different music platforms such as Spotify, Amazon Music etc. These machine learning (ML) models used user preference, listening, their history, and their behaviour to build their automatic music recommendation systems. These models also utilized the provided data set such as music artist, music user interaction, lyrics, etc to predict their recommended music (Tupe et al., 2024)

AI based deep learning (DL) models were also utilized to build their recommendation. DL models have the facilities to make a model which extracts the important features as well as to enhance the accuracy of prediction or recommendation. (Hou, 2024) utilized the DL based convolutional neural networks (CNNs) model, for a personalized music recommendation system. The CNNs model can build an efficient, personalized, user experience, and data-driven based music recommendation system with less error rate. They compared their proposed CNNs model with other ML models like support vector machine (SVM), K-nearest Neighbor (KNN), random forest (RF), and decision tree (DT). Accuracy rate, sensitivity rate, specificity rate, f1-score, and Matthews's correlation coefficients (MCC) were used to evaluate the performance of their proposed model. The high rate of accuracy rate, precision rate, recall rate, and f1-score suggest the better performance of the proposed model and lower values indicate the poor performance of the proposed model. Their proposed model achieved a 95.70% accuracy rate, 91.3% specificity rate, 97.1% sensitivity rate, 95% f-score rate, and 0.85 MCC rate in building an automatic music recommendation system. All the criteria achieved a good measurable rate which suggests the efficiency, worth, and power of the built music recommendation system.

(Reddy et al., 2024) also used the DL based approach to build a personalized and fascinating music recommendations system for music companies. They utilized a novel Gravitational Search Optimized Recursive Neural Networks (GS-RNN) approach to track the recommended music for the users. This model was a hybrid of the Gravitational Search Optimized and Recursive Neural Networks features. It was used to assess the audio signal similarity to build an automatic music recommendation system. They used accuracy rate, precision rate, recall rate, f1-score, and logarithmic loss to evaluate the performance of their proposed model. The models high accuracy rate, precision rate, recall rate, and f1-score suggest the better performance of the proposed model and lower values indicate the poor performance of the proposed model.

The higher rate of logarithmic loss indicates that the proposed model has poor performance and a poorly built automatic music recommendation system. Their proposed approach achieved an 80% accuracy rate, 84% precision rate, 88% f-score rate, 83% recall rate, and 0.85 loss rate. Furthermore, that mode also can suggest songs within the same genre.

(Bevec, et al., 2024) suggest, that users need a method that can be utilized on their application that can directly navigate their recommended songs in large collections. Collaborative filtering (CF) is a method that influences the interactions of past user items successfully but with some limitations like poor performance among sparsely connected items. They used DL based Graph neural networks (GNN) in the same content to build an automatic music recommendation system. They used the Spotify dataset against the CF method with the integration of the GNN model. This hybrid approach was utilized to enhance the performance of the automatic music recommendation system. Their proposed approach achieved venturing beyond the accuracy rate according to their experiment. They suggest the CF method doesn't show any kind of performance to drop in the long tail. The hybrid approach that was utilized by did not offer an advantage.

(Moghal Karishma, 2024) also tried to build an automatic music recommendation system using the ML approach. They suggest that the music recommendation system is a musical friend that was used to understand the user preference against their songs. They also explore different kinds

of methods already adopted by past studies to build an automatic music recommendation system. Some used filtering methods, content filtering methods, and hybrid models for music systems for users. In their study, they used the KNN model and integration of the Weighted Product Method with the KNN model. They also developed a hybrid model that will use the input of the song description and provide the different recommended songs for the user. Other methods like factorization machines, a hybrid of factorization machine and singular value decomposition (SVD), KNN, and SVD were required to make an automatic music recommendation system. They used a music dataset from 2000 to 2009 that was downloaded from the Kaggle website then they adopted some data cleaning methods to enhance the performance of the music data like removing blank and duplicating entries. Also, they discarded punctuations from the lyrics. The hybrid of KNN and Weighted Product Method achieved an 83.65% accuracy rate, the factorization machine achieved a 74.82% accuracy rate, and the hybrid model of the factorization machine and SVD model achieved a 74.84% accuracy rate. Overall, the performance of the hybrid of KNN and Weighted Product Method was greater than the other models.

(Raychaudhuri, 2024) also used an ML based algorithm to improve user interaction and based on their interaction songs will be suggested or recommended to users. Their aim in the study was to build a music recommendation engine using machine learning. User preference in songs play an essential role in increasing satisfaction levels on using music platforms. It's crucial to create a reliable and competent predictive model that increases user satisfaction towards the music platform. The predictive model will use the information or history of user interest and recommend the next song based on their choice or taste using the ML models or hybrid approach. Different states were used for the data collection process because good data builds a good predictive model. This model utilised users listening history, song metadata, audio features, and lyrics make good predictive models. The results suggest a music recommender system has achieved great outcomes from ML, DL, and hybrid approaches. Standard evaluation measurements were used for their ML models and hybrid approaches such as precision, recall, and f1-score which revealed that ML models play a key role in building a better and more efficient

system. (Alyza et al., 2023) also used the ML models to build an automatic music recommendation system.

They tried to predict the song genres using the KNN and SVM models using principle component analysis and without using principle component analysis (PCA). Mel-Frequency Cepstral Coefficients (MFCC) method was used to collect the dataset from data sources because each track used the MFCC features. 10 classes from the GTZAN dataset were used as follows: pop, reggae, metal, jazz, blues, disco, classical, hip hop, rock, and country songs for their predictive models. They used a balanced audio dataset because each class had 1000 class labels. They calculate the confusion matrix to check the correct prediction ability of ML models and their accuracy rate for comparison purposes. They found that the simple PCA method achieved a 58% accuracy rate. The efficiency of the PCA model was decreased, but the KNN model achieved a 64.9% accuracy rate and the SVM model achieved a 77% accuracy rate. However, they also built a model based on cosine similarity for an automatic music recommendation system using the same dataset. However, the cosine similarity method achieved an 80% accuracy rate which is high. The cosine similarity method efficiently recommends the songs based on their audio features.

(Hsin, 2023) also used a ML based models to build effective modern-day music recommendation. GTZAN dataset was also used for the predictive model. He utilized the KNN, SVM as ML based model, and CNN as DL based model, and used the feature extraction method to extract the important features from the music dataset. He also calculated the confusion matrix to check the correct prediction ability of ML models and DL models and their accuracy rate for comparison. The SVM model achieved an 89% accuracy rate using the Kaggle dataset. The KNN model based on the ML approach achieved a 92% accuracy rate using the aforementioned dataset. The shocking accuracy obtained from the DL based CNN model was only 62%.

(Anand, et al., 2021) used the DL based CNN and Recurrent Neural Network (RNN) models to build the music recommendation system. They used these models and compared them with previous studies in the accuracy rate to obtain a better system. The aim of the study was to make the best and most suitable predictive model that gives the song based on their tastes, characteristics, and the music they had previously heard. They used accuracy rate, precision rate, recall rate, and f1-score to evaluate the performance of their proposed model. The high rates and scores suggest the better performance of the proposed model and lower values indicate the poor performance of the proposed model. Their proposed CNN model achieved 96.60% accuracy rate, 99.3% f-score, 98.3% precision rate, and 98.3% recall rate. The DL based, RNN model achieved an 84.4% accuracy rate using the 20 number of Epochs or iterations. Comparatively, the RNN model achieved an 84.30 recall rate, 85% precision rate, and 85.5% f-score rate. However, the performance of the RNN model was not achieved better as compared to the CNN model, which suggests that DL based CNN model provides better performance to build an automatic music recommendation system.

(CH et al., 2021) also build their ML based with music recommendation system with advanced classification which used SVM, KNN, and decision tree ML models for their system. A training testing strategy was needed to split the data into two groups. The training data was used to train the SVM, KNN, and decision tree models, and testing data was used to test the SVM, KNN, and decision tree models. The SVM model achieved only a 40.92% accuracy rate for testing data and a 42.96% accuracy rate for training datasets. The KNN model achieved only a 57.32% accuracy rate for testing data and a 99.32% accuracy rate for training datasets. The decision tree model achieved only a 57.32% accuracy rate for testing data and a 99.32% accuracy rate for training datasets.

The SVM model achieved too many poor or underfitting results to build an automatic music recommendation system. The KNN and decision tree models achieved a high accuracy rate for the training dataset but too low for the testing dataset. The KNN and decision tree models have overfitting issues which reduce the ability of an automatic music recommendation system while

using the unknown testing dataset. They also suggest that future research will add more or other music features to enhance the accuracy rate of predictive models. The improved accuracy of the models also improved the ability of an automatic music recommendation system.

3 Methodology

3.1 Data and software description

The 30000-spotify-songs were used for this study analysis which is downloaded from the Kaggle website (www.kaggle.com, n.d.). The (track id, track name, track artist, track popularity, track album id, track album name, track album release date, playlist name, playlist id, playlist genre, playlist subgenre, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration in ms) all were the variables included in the data. In total, 23 variables were part of the data and this study. The (track id, track name, track artist, track popularity, track album id, track album name, track album release date, playlist name, playlist id, playlist genre, playlist subgenre, and mode) were the categorical variables in selected data.

(danceability, energy, key, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration in ms) were the continuous variables in selected variables. To fulfill the study objective of building an AI-Based Music Recommendation System, Python was used. The Python environment, Jupyter Notebook was preferred for this study analysis (jupyter, 2019). It allows me to perform the K-Means cluster analysis, and KNN machine learning model analysis.

Exploratory data analysis (Data preprocessing and partition)

The imported data will be preprocessed using standard methods like handling missing values, outliers, and repeated values using the exploratory data analysis. The exploratory data analysis provides the facility to understand the data trend, pattern, help in detect missing values, outliers, relationship between the variables, features engineering, summarize and visualize the data, and provide better decision about data (GeeksForGeeks, 2018).

So the most relevant features related to the music system will be extracted for the best AI-based system. The extracted features will be used in the further transformation process. The “StandardScaler()” method will be used to transform (Scikit-learn.org, 2019) the important features for better cluster analysis and better prediction based on the KNN model. The training test method was also utilized to split the dataset into training and testing parts. The

“train_test_split” function will be used for data partition, the 70% whole data will be split as training samples and 30% will be split as testing samples (scikit-learn, 2018). The training part was used to train the proposed model, and testing was used to test the model.

3.2 Models

3.2.1 Cluster Analysis

The k-means cluster analysis will be used to cluster the data. The cluster analysis will make the cluster of the data using their similarity or similar information or characteristics. The similarity will be determined using the Euclidean distance, cosine similarity, etc (scikit-learn, 2010). In this study, it will play an important role in making the AI-Based Music Recommendation System. The cluster analysis will make the cluster of selected features on the base of their similar characteristics. Next the same characteristics further will be used for the detection process using the distance method. Which characteristics are close to a particular cluster these features lie in that cluster.

3.2.2 K-Nearest Neighbor

The KNN model is a machine learning model that is used for regression prediction and classification processes. The KNN model will be used in this study to predict the important features of the music of 30000 Spotify Songs. The KNN model will be trained on the training data and tested on the testing dataset (Scikit-learn.org, 2019a). The KNN model also calculates the similarity between the features based on their distances. The K (number of neighbors) is the parameter of the KNN model that was used. It's also crucial to affect the results of the KNN Model due to the wrong selection of the K number. Too many small values make the algorithm sensitive and causes prediction errors and noise (GeeksforGeeks, 2019). In the same way, large values also affect the KNN model. It will optimize the performance of the AI-Based Music Recommendation System to recommend the most relevant songs to users. It can also be used for image recognition process based on the image's pixels or intensities. This can also be used in the medical field to diagnose the patient subjects on the base of their condition attributes.

3.2.3 AI-based Automatic music recommendation system

After extraction of the suitable clusters from the K-means cluster analysis, the KNN will be used to make the AI-Based Music Recommendation System. Both K-means and KNN approaches make an efficient AI-Based Music Recommendation System. The integration of the K-means with the KNN model improves the performance of the AI-Based Music Recommendation System using the 30000-spotify-songs data. The purpose is to improve the AI-Based Music Recommendation System with lower error rates. The results of the KNN model based on the cluster analysis will be measured using the mean square error (MSE), root mean square error (RMSE) and mean absolute error (MAE).

3.2.4 Important libraries

The important libraries of Python will provide facilities for data importing, data preprocessing, data recording, data clustering, and AI-based music Recommendation Systems. The description of the important libraries is given in Figure 1. The (pandas, numpy, seaborn, matplotlib, and sklearn) are the libraries used in this study to import the data, preprocess the data, plot the data, and analyze the data using the Python Jupyter Notebook. Pandas is a library used to handle data frame such as CSV, and SQL and make them easy to handle and analyze. Numpy is usually used for multi-dimensional arrays and matrices, or it can be used for scientific computing. Seaborn is used for the visual display of data for graphs and charts, matplotlib is also used for visualization including static and animations; it can also customize a wide variety of plots and graphs, scikit-learn is used for machine learning for knowledge discovery in the database and its analysis using a diversity of algorithms like regression, cluster analysis, and KNN. The principal component analysis will also be performed using the sklearn package in jupyter notebook.

```
import pandas as pd
import numpy as np
# Data visualization
import seaborn as sns
import matplotlib.pyplot as plt
# Machine Learning
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
```

Figure 1: Important libraries for data analysis.

4 Implementation

4.1 Exploratory Data Analysis

To fulfil the study objectives the data was imported into Jupyter Notebook. The “pd.read_csv” function was used to import the “Spotify songs.csv” data in Jupyter Notebook. The “data.head” function was used to print the first 3 rows with all 23 variables are given in Figure 2. There was a total of 23 variables and each variable had 32833 observations.

```
# Database
data = pd.read_csv("E:/data/spotify_songs.csv")
data.head(3)
```

	track_id	track_name	track_artist	track_popularity	track_album_id	track_album_name	track_album_release_date	playlist_name
0	6f807x0lma9a1j3VPbc7VN	I Don't Care (with Justin Bieber) - Loud Luxur...	Ed Sheeran	66	2oCs0DGTsRO98Gh5ZSI2Cx	I Don't Care (with Justin Bieber) [Loud Luxur...	2019-06-14	Pop Remix
1	0r7CVbZTWZgbTCYdfa2P31	Memories - Dillon Francis Remix	Maroon 5	67	63rPSO264uRjW1X5E6cWW6	Memories (Dillon Francis Remix)	2019-12-13	Pop Remix
2	1z1Hg7Vb0AhHDIEmnDE79l	All the Time - Don Diablo Remix	Zara Larsson	70	1HoSmj2eLcsrR0vE9gThr4	All the Time (Don Diablo Remix)	2019-07-05	Pop Remix

3 rows × 23 columns

data.shape
(32833, 23)

Figure 2: Imported data.

4.1.1 Data Preprocessing and Partition

The values that were initially missing were checked for imported data, and the existing missing values from data were removed. The missing values may lead to wrong information from selected data; sometimes, missing values disturb the descriptive statistics measure of the continuous variables. This also affects the other statistical analyses, so it's important to check and remove the missing value issue in the data. After removing the missing values from the imported data, the remaining data shape was (32828, 23). Only 5 values were removed from the data to make the data cleaner, and detail is given in Figure 3. Figure 3 also suggests that there are no duplicate rows existing in the cleaned dataset. The training and testing split method splits the data into training parts then the 11 most relevant variables were selected to build an AI-based automatic music recommendation system. The 22979 observations were divided into the training part, and 9849 observations were split into the testing part out of 32828 observations.

```
data.dropna(inplace=True)
data.isnull().sum(), data.shape, print(f"Number of duplicate rows: {data.duplicated().sum()}")
Number of duplicate rows: 0
(track_id          0
 track_name        0
 track_artist      0
 track_popularity  0
 track_album_id    0
 track_album_name  0
 track_album_release_date  0
 playlist_name     0
 playlist_id       0
 playlist_genre    0
 playlist_subgenre  0
 danceability      0
 energy            0
 key               0
 loudness          0
 mode              0
 speechiness       0
 acousticness      0
 instrumentalness  0
 liveness          0
 valence           0
 tempo            0
 duration_ms       0
 dtype: int64,
 (32828, 23),
 None)
```

Figure 3: Missing values handling.

4.1.2 Bar plot and Correlation Analysis plot

Exploratory data analysis was used to explore the data, where the graphical presentation of the codes to plot the “playlist genre” is given in Figure 4. These codes will produce the bar charts for each category of the playlist genre”. The “seaborn” library was used for graphical presentation of the variables. The “sns.countplot” function will produce the barplot for the playlist genre” variable.

```
# Count of tracks by playlist genre
sns.countplot(data, x='playlist_genre')
plt.title('Bar plot of Tracks by Playlist Genre')
plt.xticks()
plt.show()
```

Figure 4: Codes to make a Bar plot.

The correlation heatmap of the continuous variables was also drawn to observe the relationship between the continuous variables. The Pearson correlation will be used to measure the correlation between the variables. The implementation codes to make the correlation plot for continuous variables is given in Figure 5. The (track popularity, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration in ms) variables were used for the correlation plot.

```
# Selecting only numeric columns
numeric_df = data.select_dtypes(include=['float64', 'int64'])
plt.figure(figsize=(10, 6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Plot')
plt.show()
```

Figure 5: Correlation plot codes.

The detailed description of Pearson's correlation ranges is given in Table 1 for the positive and negative relationship between the variables. The positive correlation means the pair of two independent variables are move in the same direction. The same direction suggests that as much as values or quantity in one variable increase, then the second variable also increases. Both move in the same positive direction. The negative correlation means the pair of two independent variables don't move in the same direction; these variables move in the opposite direction. The opposite direction suggests that when the values or quantity in one variable increase then the second variable will decrease, and both move in the opposite direction.

Table 1: Evaluation report for Pearson's correlation.

Interval Coefficient of Pearson's correlation	Level of relationship	Interval Coefficient of Pearson's correlation	Level of relationship
0.80 to 1	Very strong positive	-0.80 to -1	Very strong negative
0.60 to 0.79	Strong positive	-0.60 to -0.79	Strong negative
0.40 0.59	Moderate positive	-0.40 -0.59	Moderate negative
0.20 to 0.39	Weak positive	-0.20 to -0.39	Weak negative
0 to 0.19	Very Weak Positive	-0 to -0.19	Very Weak negative

4.2 Cluster Analysis

The K-Means cluster analysis was implemented with extracted features such as ('danceability', 'energy', 'key', 'loudness', 'mode', 'speechiness', 'acousticness', 'instrumentalness', 'liveness', 'valence', 'tempo'). These are the most relevant features of songs. The implementation codes for the K-Means cluster analysis are given in Figure 6. In cluster analysis we used the following codes in firstly k (cluster length) is selected using the range command, then a for loop is used to fit on

the train the data. In the other half simply use the figure command to plot the results for the x and y axes.

```
inertia = []
k_range = range(2, 11)
for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_train)
    inertia.append(kmeans.inertia_)
# Plot Elbow Method graph
plt.figure(figsize=(8,5))
plt.plot(k_range, inertia, marker='o', linestyle='--',color="red")
plt.xlabel("K")
plt.ylabel("Inertia")
plt.title("Elbow Method for Optimal K")
plt.show()
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
data['Cluster'] = kmeans.fit_predict(df_scaled)
```

Figure 6: K-Means implementation codes.

4.3 KNN model

The KNN model was trained on the training data with Euclidean distance and 6 n_ neighbors. The recommend_songs function was created to test the performance of the KNN model. The detailed implementation codes of the KNN model and recommend_songs function are given in Figure 7. The features will be used in recommend_songs and these features will be reshaped to process the calculation of the distances and indices using the KNN model. Next based on KNN and cluster analysis, the different song's names will be imported into the function that will be used in the KNN model and predict the song's distances. These distances will provide the facility to check the songs in the different clusters on the base of similarity and distances. The "Memories - Dillon Francis Remix" song was used to test the recommended 5 songs against the "Memories - Dillon Francis Remix" song.

```
knn = NearestNeighbors(n_neighbors=6, metric='euclidean')
knn.fit(X_train)
def recommend_songs(song_name, data, features, scaler, knn):
    song_idx = data[data['track_name'] == song_name].index[0]
    song_features = df_scaled.iloc[song_idx].values.reshape(1, -1)
    distances, indices = knn.kneighbors(song_features)
    recommended_songs = [data.iloc[idx]['track_name'] for idx in indices[0][1:]]
    plt.figure(figsize=(6, 3))
    plt.barh(recommended_songs[:-1], distances[0][1:][:-1], color='grey')
    plt.xlabel("Similarity Distance")
    plt.title(f"Top 5 Similar Songs to '{song_name}'")
    plt.gca().invert_yaxis()
    plt.show()
# Example usage
recommend_songs("Memories - Dillon Francis Remix", data, X_test, scaler, knn)
```

Figure 7: KNN implementation codes to make the music recommendation system.

5 Results

5.1 Exploratory Data Analysis

To fulfil the study objectives, the data was imported into Jupyter Notebook. This section will provide a description or explanation of the results related to the analysis. The imported data with its first 3 rows with all 23 variables. There was a total of 23 variables and each variable had 32833 observations.

5.1.1 Data Preprocessing

The preprocessing step is necessary in data analysis, initially missing values were checked for imported data and the existing missing values from the data were removed. After removing missing values from the imported data, the remaining data shape was (32828, 23). Only 5 values were removed from the data to make the data cleaner. The most relevant variables related to songs were extracted as important features for the AI-based automatic music recommendation system. The extracted features will be used in the further transformation process. The “StandardScaler()” method will be used to transform the important features for better cluster analysis and better prediction based on the KNN model. The Standard Scaler method was applied using the scikit-learn method and it standardized the important features by removing the mean and scaling to unit variance. The Exploratory Data Analysis was used to explore the data, where the graphical presentation of the playlist_genre in the form of a bar plot with blue color is given in Figure 8. In the dataset, there are six types of playlist_genres shown in the selected dataset such as pop, rap, rock, Latin, r&b, and EDM. Rap, pop, and r&b categories have the most playlists.

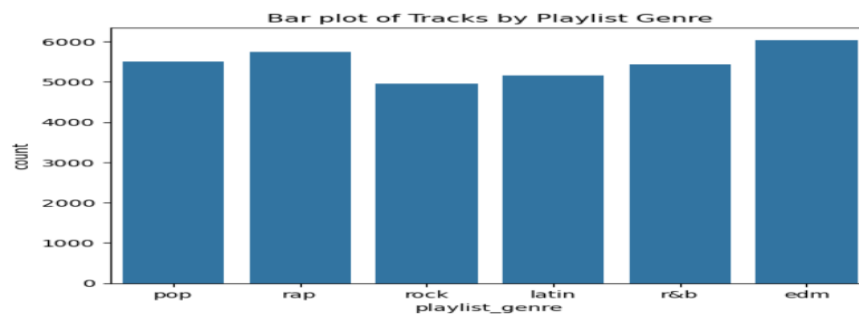


Figure 8: Bar plot of playlist_genre variable.

5.1.2 Correlation Analysis

The correlation heatmap of the continuous variables is given in Figure 9. The heatmap presents the correlation between the variables, with blue presenting the negative correlation between the variables and white to red presenting the positive correlation between the variables (GeeksforGeeks, 2020). The close-to-zero value suggests no correlation between the variables. A close to 1 means a high positive correlation exists between the variables and close to -1 suggests a high negative correlation between the variables. The diagonal red color presents the correlation or relationship of the variable with itself that will be always 1. The off-diagonal values present the pairwise correlation between the variables. The above and lower off-diagonal values were showing the same results or same correlation values between the pairs. We can use above or lower off diagonal values for interpretation. The energy, instrumentals, and duration in ms have a negative correlation with track popularity which suggests that as much as energy, instrumentals, and duration in ms increase, track popularity will decrease the remaining variables have a poor correlation with track popularity.

Speechiness and valence have a positive correlation with danceability, and liveness and tempo have a negative correlation with danceability. The negative correlation means both have opposite directions and the positive means both have the same direction. Its loudness has a moderate positive correlation with energy, and liveness, valence, and tempo have a small correlation with energy all moving in the same direction. But acoustics have a moderate negative correlation with energy both move in opposite directions.

The remaining do not correlate with the energy variable. Acoustics, instrumentals, and duration in ms have a small negative correlation with loudness which means its all moving in the opposite direction. The remaining do not correlate with the loudness variable. Instrumentals have a small negative correlation with valence which means both move in the opposite direction. But the liveness variable has no significant correlation with valence, tempo, and duration in ms variables.

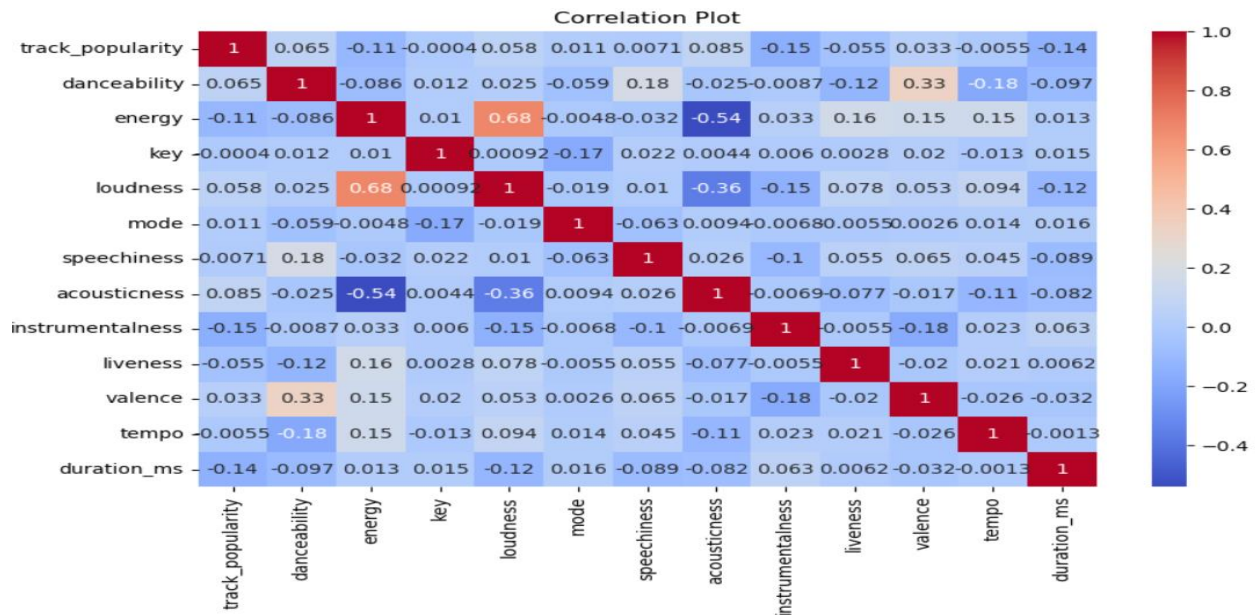


Figure 9: Correlation plot between continuous variables.

5.2 Cluster Analysis

The K-Means cluster analysis was implemented with extracted features such as ('danceability', 'energy', 'key', 'loudness', 'mode', 'speechiness', 'acousticness', 'instrumentalness', 'liveness', 'valence', 'tempo'). The Elbow method was mostly used in K-Means clustering analysis to determine the best number of clusters or K. The graphical presentation of the Elbow method with

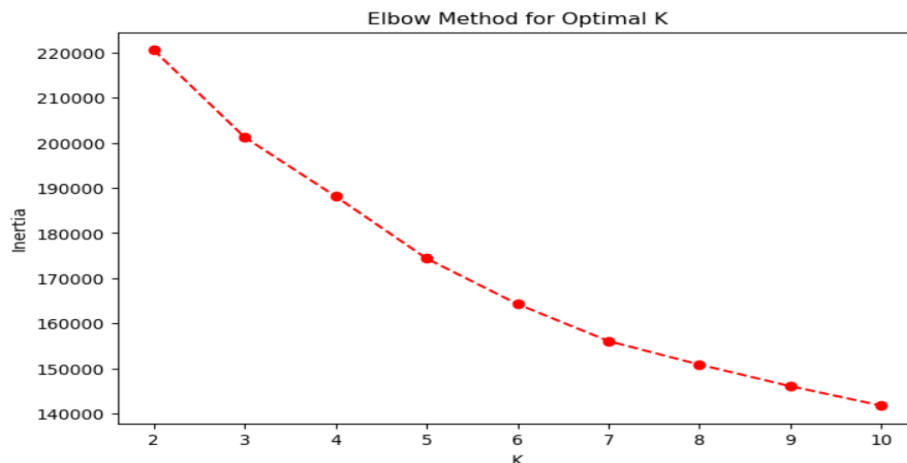


Figure 10: K-Means cluster analysis for select features.

10 numbers of K neighbors are given in Figure 10. Where the X-axis presents the number of clusters or K, and the Y-axis presents the sum squared distances between the values or points against the number of clusters (K). The red dashed line with “O” marker was showing the number of clusters. It is challenging part to decide the correct number of K for K-Means cluster analysis. Because above 7 or higher values of K may produce overfitting issues which leads to too many small clusters. Similarly, lower than 3 values of K may produce the underfitting issues which leads to too many clusters. As much as K increases and it shows a decreasing pattern it suggests the best fitting of the cluster analysis. In this case, we can observe the graphical presentation of K-Means cluster analysis in Figure 10.

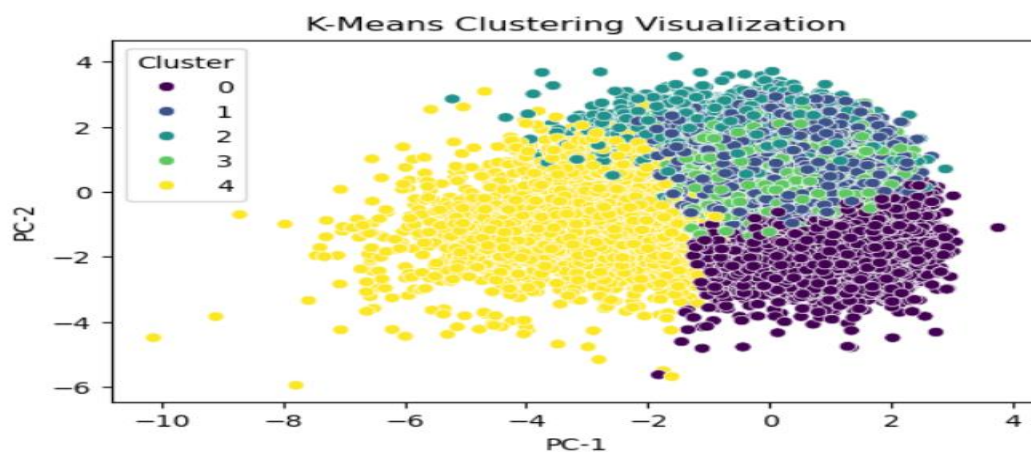


Figure 11: K-Means cluster analysis visualization of 5 clusters.

The principal component analysis was performed in cluster analysis and a scatter plot was used to plot the principal components of the cluster analysis against the 5 clusters (GeeksforGeeks, 2018). Here we choose the 5 number of clusters to prevent overfitting and under-fitting issues. The extracted or selected features will be grouped into 5 clusters on the base of similarity. The purple color presenting the cluster 0, blue color presenting the cluster 1, dark green color presenting the cluster 2, light green color presenting the cluster 3, and yellow color presenting the cluster 4. The PCA will reduce the dimension of the data into 2 principal components such as PC1 and PC2. The graphical presentation of the PC1 and PC2 is given in Figure 11. Where the X-axis presents the PC1, which represents the most significant feature combination that contributes to variance in the data. Similarly, the Y-axis presents the second most significant feature combination that contributes to variance in the data for the music recommendation

system. In Figure 10 we can observe the songs were grouped into 5 clusters based on their similarities.

5.3 KNN Model and Testing

The KNN model was trained on the training data with Euclidean distance and test and testing data. The KNN model achieved a 1.15 MSE value for training features, a 1.07 value RMSE for training data, and a 0.88 MAE value from the training dataset. Similarly, the KNN model achieved a 1.46 MSE value for testing features, a 1.21 value of RMSE for testing data, and a 1.10 MAE value from the testing dataset. The lower values of MSE, RMSE, and MAE for training and testing data indicate the good performance of the KNN model. A detailed description of the training and testing dataset is given in Table 2.

Table 2: Evaluation report using the KNN model for training and testing data.

Data	Mean Squared Error	Root Mean Squared Error	Mean Absolute Error
Training	1.15	1.07	0.88
Testing	1.46	1.21	1.10

After measuring the performance of the KNN model on training and testing data at the end the builded function was tested on song. The recommend_songs function was tested on testing dataset to measure the performance. The “All the Time - Don Diablo Remix” song was used in the recommend_songs function which predicts the 5 similar songs for “recommend_songs function “using the distance method of cluster analysis. The AI-based music recommendation system suggests the (Highest in the room, wow, the less I know the better, nann n****, All the Time) 5 songs related to the “All the Time - Don Diablo Remix” song. The graphical presentation of the recommend_songs function on the testing dataset is given in Figure 12. The distance will provide information about the similarity of the song, a high distance means a high accuracy rate and the lower distance means a lower accuracy.

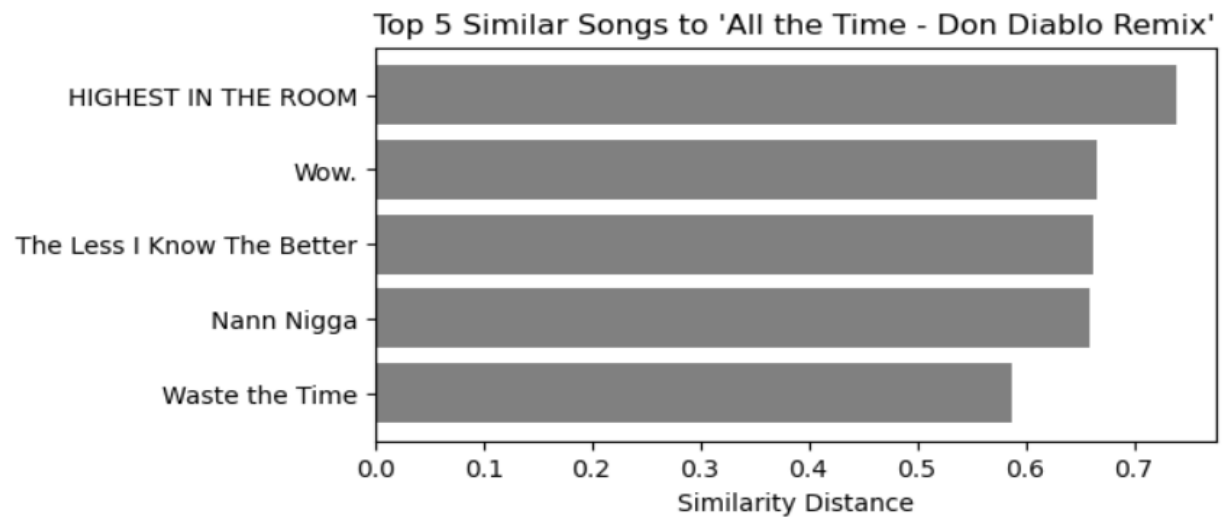


Figure 12: Prediction of the 5 songs using the `recommend_songs` function.

6 Evaluations

An AI-Based Music Recommendation System development was done using K Means Clustering and K Nearest Neighbors (KNN) to make an efficient and individualized recommendation engine, adding value to the user experience by presenting similar song suggestions. The project was able to apply machine learning techniques to classify and recommend songs to users, in a way that accounts for major preprocessing challenges, model optimization, and computational efficiency and user feedback integration.

The project evaluation covered both technical outcomes and implementation obstacles together with responses to the problems and suggestions for future system enhancements. The system effectiveness measurement relied on established industry metrics including Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) to assess prediction accuracy for user preferences. The real music data testing of the recommendation function confirmed that the system effectively clustered songs along with generating quality recommendations.

This part provides an extensive project assessment through the description of encountered problems and implemented solutions as well as testing results and upcoming strategies to enhance the AI-based music recommendation framework.

6.1 Challenges and solutions

The project faced problems with data processing and performance plus processing efficiencies. Differential Decision Support system development faced challenges through this process that required data management performance updates.

6.2 Data Preprocessing and Feature Extraction

The main difficulty in creating the music suggestion system stems from processing a large volume of unlabeled data. Music files in digital format include various data properties such as audio quality metrics plus information about tracks and user interactions. The work made data processing speed essential and worked with high standards of data quality.

- This project needed feature engineering plus data cleaning processes to work properly.

- The processing steps of normalization and standardization helped us normalize all music features evenly.
- Using interpolation and imputation methods to fill incomplete data employs methods that keep recommendations neutral and unbiased.
- PCA analysis sorted through music attributes to pick the most important elements which cut system processing needs while keeping results reliable.

These preparation steps made the dataset more practical for use in ML models that could suggest reliable results.

6.3 Model Performance Optimization

The first implementation of KNN model had performance issues because finding the nearest neighbors in large datasets is very expensive computationally. KNN is a lazy learning algorithm, so it involves considerably high computational power for finding out the distance between the data points which often slows down its response time.

Solution to the problems above were:

- Hyperparameter Tuning: To fine tune the accuracy and reduce the computational load, different value of K (number of neighbors) was experimented on.
- Depending upon the type of data distribution, it was observed that the accuracy could be improved by using Manhattan and Minkowski instead of Euclidean distance metrics only.
- KNN searches were significantly speeded up from used of KD-Tree and Ball-Tree Indexing, and as a result recommendation processes were improved.

With these improvements, the model was becoming faster and more accurate for making recommendations, and the model optimization strategies were proving to be effective.

6.4 Testing outcomes

Data	Mean Squared Error (MSE)	Root Mean Squared Error (RMSE)	Mean Absolute Error (MAE)
Training	1.15	1.07	0.88
Testing	1.46	1.21	1.10

Overall, the anchors are highly accurate in terms of creating context for similar songs based on the very low MSE, RMSE, and MAE co-efficients. The difference between the training and test set error was low, which will help to achieve good generality.

Regarding the functions, the recommend_songs function was checked with the input song “All the Time - Don Diablo Remix” and the system came up with five related songs with a distance metrics. The findings thus supported the ability of the implemented AI model to cluster and suggest relevant music that had a high level of similarity.

6.5 Alignment of testing outcomes with objectives

6.5.1 Objective 1: AI-Based Music Recommendation System Development

- The system achieved success by processing Spotify song data with machine learning models to suggest appropriate songs.
- The test showed this system works properly by recommending five relevant tracks from the test song "All the Time - Don Diablo Remix".

6.5.2 Objective 2: K-Means Clustering for Music Categorization

- Using K-Means clustering the algorithm put songs into five groups through their shared audio traits danceability, loudness, and tempo.
- Both clustering and correlation tests showed which musical characteristics affected which type of music and made popular songs.

6.5.3 Objective 3: KNN Model Performance for Recommendations

- The KNN system showed these testing results during its evaluation phase.
 - The system calculated mean squared error as 1.15 and root mean squared error as 1.07 while maintaining 0.88 mean absolute error during training sessions.
 - Testing the model yielded MSE 1.46 and RMSE values equal to 1.21 while MAE stood at 1.10.
- The model shows its ability to predict good quality song suggestions with new data by its small error results.

All predetermined objectives were met through the implementation of AI-based music recommendation techniques by the system. Testing results show that KNN model produces

minimal errors and establishes its reliability as an accurate classifier while K-Means clustering performs efficiently in music category assignment. Future development including real-time learning alongside deep learning integration would improve system adaptability and accuracy level. The system achieves its stated objectives and establishes a robust basis for AI-based music recommendation systems because of current testing results.

6.6 Recommendations

- The proposed model based on cluster analysis and the KNN model can recommend more songs to the users of Spotify.
- The AI-based models also gained significant popularity in different streaming services. These models provide facilities for an automatic music recommendation system for different music platforms such as Amazon Music, Soundcloud, Spotify, etc.
- This strategy provides the help to build a personalized and fascinating music recommendations system for music companies.
- The model is prone to attracting new users and satisfying the current users.
- By utilizing the proposed strategy users can directly navigate their recommended songs in large collections.
- After implementation of the provided strategy user can experience a more memorable experience for their singing test.
- The improved experience of current users who recommend the music platform to other non-users, leads to an increase in the audience.
- It uses a hybrid approach of Cluster analysis and KNN model which provides a minimum error rate without any over-fitting and under-fitting. So, this can be profitable to the company, the users will be more comfortable and satisfied if they get songs according to their tastes, and preferences.
- This model can also be used by other companies as it can increase Spotify users, and they need such an approach to increase their approach to users. If this model is used it will increase both user attraction and satisfaction.
- This modal can provide real-time five recommended songs to the users in their playlist.

- The danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentality, liveness, valence, and tempo were the important features that can be used to increase the attention against the music company. Companies should focus on these important key features.

6.7 Future Recommendations

- The song recommendation system can be improved using lyrical sentiment analysis such as happy lyrics, sad, energetic, and others to improve the feature selection. Similarly, user behavior data against the songs like their skipping rate, their number of repetitions, and number of likes against the playing songs.
- The other different methods for feature selection or dimension reduction such as the random forest method to retrieve more important features to build an AI-based automated music recommendation system.
- We can also use other clustering methods like auto encoders, and neural collaborative filtering to learn the song pattern in more detail for a better music recommendation system.
- The hybrid recommendation model can also be used to enhance the model performance such as a combination of the content-based filtering and collaborative filtering method and use other methods such as matrix factorization for better user-song relationships.
- In this case the Euclidean distance, method was used for the KNN model we can try other such as cosine similarity and dynamic time-warping distance methods.
- The deep learning model can also be used to improve the performance of songs or music recommendation systems such as a one-dimensional convolution neural network (CNN), and recurrent neural network (RNN).
- We can also use the hybrid of the machine learning and deep learning model to improve the model evaluation metrics such as MSE, RMSE, and MAEs.
- Also, other metrics such as precision, recall, and mean reciprocal rank to measure the performance of the music recommendation system.

7 Conclusion

The study was designed to build the AI- Based Music Recommendation System, apply K-means cluster analysis and KNN model to enhance the Music Recommendation System. To fulfil the study objectives the Spotify songs data was imported into Jupyter Notebook. There was a total of 23 variables and each variable had 32833 observations and data was cleaned to remove the missing values, outliers, standardized the data, select the important music features for music recommended system. The correlation plot revealed that, energy, instrumentals, and duration in ms have a negative correlation with track popularity. Similarly, speechiness and valence have a positive correlation with danceability, and liveness and tempo have a negative correlation with danceability. Similarly, loudness has a moderate positive correlation with energy, and liveness, valence, and tempo have a small correlation with energy. But acoustics have a moderate negative correlation with energy. Acoustics, instrumentals, and duration in ms have a small negative correlation with loudness. Similarly, instrumentals have a small negative correlation, and the liveness variable have no significant correlation with valence, tempo, and duration in ms variables and same for other pairs.

The cluster analysis suggests the 5 best clusters of songs on the base of their similarity. The KNN model was trained on the training data with Euclidean distance and test and testing data. The KNN model achieved a 1.15 MSE value for training features, a 1.07 value RMSE for training data, and a 0.88 MAE value from the training dataset. Similarly, the KNN model achieved a 1.46 MSE value for testing features, a 1.21 value of RMSE for testing data, and a 1.10 MAE value from the testing dataset. The lower values of MSE, RMSE, and MAE for training and testing data indicate the good performance of the KNN model. After measuring the performance of the KNN model on training and testing data at the end the built function was tested on song. It works wonderfully on the data.

Research results demonstrate that machine learning improves music recommendation quality and user experience. K-Means helped group music successfully and the KNN model then generated customized song recommendations from user preferences. Research proves that using audio data patterns plus existing statistical relationships helps create better music recommendation services. The models use data from the musical traits and listener habits to demonstrate how real platform providers should apply this technology. Future research needs to test how deep learning approaches with combined recommendation strategies enhance the system when users change their behavior patterns.

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